

The Orbital Vanguard: A Strategic and Historical Analysis of the X-37B and the Evolution of Military Spaceplanes

The realm of military aerospace has long been defined by a fundamental dichotomy: the pursuit of ultimate high-ground advantage versus the rigid constraints of orbital mechanics and terrestrial economics. For decades, spacecraft were largely expendable, bespoke assets relegated to predictable trajectories, while atmospheric aircraft offered reusability and maneuverability but remained inextricably bound by the atmosphere. The synthesis of these two domains—the spaceplane—represents one of the most complex engineering challenges in modern history. At the forefront of this technological convergence is the Boeing X-37B Orbital Test Vehicle (OTV), a classified, uncrewed, reusable spacecraft operated by the United States Space Force.

Since its inaugural orbital flight in 2010, the X-37B has become an emblem of American space dominance, a catalyst for geopolitical anxiety, and the cornerstone of the burgeoning doctrine known as Dynamic Space Operations. Yet, the X-37B is not an isolated anomaly; it is the culmination of nearly a century of theoretical physics, Cold War espionage, and iterative materials science. As peer competitors like China field their own spaceplanes and commercial entities pursue single-stage-to-orbit architectures, the orbital environment is rapidly transitioning from a static theater of observation to a dynamic domain of maneuver. This report provides an exhaustive analysis of the X-37B program, its deep historical lineage, the technological systems that enable its endurance, the geopolitical controversies surrounding it, and the contemporary proliferation of spaceplane technologies.

Part I: The Morphological Ancestry of the Spaceplane

The architectural DNA of modern spaceplanes can be traced directly back to the early theoretical frameworks of the Cold War and the decades preceding it. The foundational concept underpinning the spaceplane is the lifting body—a vehicle that generates aerodynamic lift via the shape of its fuselage rather than traditional wings. This approach originated from the sheer necessity of surviving the brutal thermal and aerodynamic forces of atmospheric reentry, where traditional wings would be instantly incinerated or sheared off.

From Silbervogel to Dyna-Soar

The earliest conceptualization of a lifting-body spaceplane was Eugen Sänger's *Silbervogel* (Silver Bird) in the mid-1930s. Proposed as a rocket-powered sub-orbital bomber for Nazi Germany, the *Silbervogel* featured a flattened fuselage and cropped wings designed to skip along the upper stratosphere, thereby exponentially extending its range¹. While it never progressed beyond wind-tunnel models, the underlying physics heavily influenced the United States Air Force's (USAF) X-20 Dyna-Soar program, which commenced in the late 1950s³.

The X-20 Dyna-Soar was conceived as a piloted, reusable military spaceplane capable of reaching orbit atop a Titan III rocket. Its operational mandates were staggering for the era: conducting high-altitude reconnaissance, performing satellite interception, executing orbital bombardment, and gliding back to a conventional runway landing³. The Dyna-Soar was elegantly conceived but astronomically complex in execution, particularly regarding thermal protection systems, which utilized heavy, exotic refractory metals³. Despite the selection of eight elite test pilots and significant hardware development, the program was abruptly canceled in December 1963 by Secretary of Defense Robert McNamara. McNamara concluded that the program lacked a definitive military objective, redirecting resources toward NASA's crewed Gemini program and expendable, automated reconnaissance satellites³.

The Lifting Body Era and Transnational Espionage

Following the demise of the X-20, NASA and the USAF pivoted toward empirical aerodynamic research through a series of heavyweight lifting body aircraft flown at the Flight Research Center at Edwards Air Force Base. Engineers recognized that the traditional capsule design used in Project Mercury was crude, offering a hypersonic lift-to-drag ratio of roughly 0.5—marginally better than a ballistic blunt body⁵. Conversely, lifting bodies patterned on a "half-cone" or "inverted airfoil" design promised cross-range maneuverability, allowing pilots to steer the craft to specific runways upon return from space⁵.

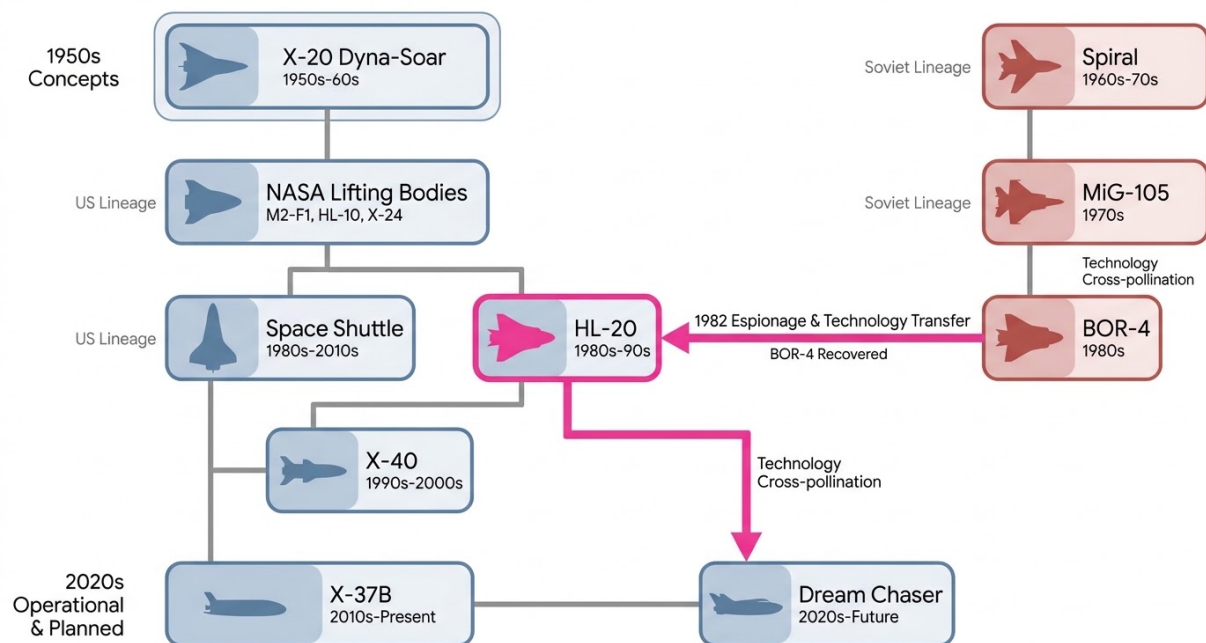
Throughout the 1960s and 1970s, legendary test pilots such as John Manke, William Dana, Jerauld Gentry, Peter Hoag, and Bruce Peterson flew a fleet of experimental vehicles, including the M2-F1, M2-F2, M2-F3, HL-10, and the X-24 series³. These vehicles were dropped from the wing of a B-52 mothership and piloted to unpowered "dead-stick" landings on the dry lakebeds³. The Northrop HL-10 proved exceptionally successful, logging 37 flights and reaching a top speed of Mach 1.86 (1,228 mph) and a peak altitude of 90,030 feet⁹. The data derived from the HL-10 and X-24A directly mitigated the aerodynamic risks of the impending Space Shuttle program⁸.

Concurrently, the Soviet Union pursued its own spaceplane ambitions under the highly classified Spiral program, initiating the development of the Mikoyan-Gurevich MiG-105 spaceplane, nicknamed "Lapot" (bast shoe) due to its blunt nose⁵. The Soviet engineering approach to Spiral was highly innovative. It utilized a "50/50" midair launch scheme, intending to launch the spaceplane and a liquid-fuel booster from the back of a custom hypersonic mothership¹². The Spiral featured variable-dihedral wings that folded upward at 60 degrees during launch and reentry for thermal protection, then lowered to a horizontal position for subsonic gliding¹². It was to land on deployable skids and utilize "scale-plate armor" made of niobium alloy and molybdenum disilicide for heat shielding¹².

To validate the thermal protection and aerodynamic stability of the Spiral design, the Soviets launched a series of subscale, uncrewed orbital test vehicles known as BOR (Unpiloted Orbital Rocketplane)⁵. This program inadvertently catalyzed the next generation of American spaceplane design through a remarkable instance of aerospace espionage. In 1982, the BOR-4 vehicle (designated Kosmos 1374) splashed down in the Indian Ocean. A Royal Australian Air

Force P-3 Orion reconnaissance aircraft covertly intercepted the Soviet recovery operation, capturing high-resolution photographs of the spacecraft¹⁴.

NASA Langley engineers extensively analyzed these classified photographs, discovering that the Soviet design—a heavily swept lifting body with a blunt nose—possessed highly efficient cross-range capabilities and excellent thermal dissipation profiles¹³. Leveraging this intelligence, NASA reverse-engineered the BOR-4 geometry into a domestic concept known as the HL-20 Personnel Launch System (PLS)². Envisioned as an emergency rescue vehicle and ferry for Space Station Freedom, the HL-20 was designed to carry ten personnel². Although the HL-20 was canceled in 1993 after years of study, its aerodynamic mold line survived for decades. SpaceDev, later acquired by the Sierra Nevada Corporation (now Sierra Space), licensed the HL-20 concept, which ultimately serves as the direct architectural blueprint for the contemporary Dream Chaser spaceplane⁵.



Part II: The 'Lost Generation' and the Pivot to Autonomous Systems

As the Space Shuttle program matured throughout the 1980s and 1990s, its immense logistical footprint, required standing army of roughly 9,000 personnel, and inherent fragility underscored the need for smaller, more efficient, and fully reusable launch vehicles⁶. During this period, NASA pursued several ambitious, billion-dollar replacements that ultimately ended in cancellation but laid the critical groundwork for the autonomous systems used today.

The most ambitious of these was the Lockheed Martin X-33, a sub-scale technology demonstrator for the proposed *VentureStar* commercial spaceplane. The X-33 attempted to achieve the highly coveted "holy grail" of aerospace engineering: Single-Stage-To-Orbit (SSTO) flight⁶. The wedge-shaped lifting body utilized highly advanced linear aerospike engines—which

used the vehicle's body as the nozzle wall to adapt automatically to changing atmospheric pressures—and lightweight composite cryogenic fuel tanks⁶. After expending \$1.28 billion and reaching 85% assembly, the program was canceled in 2001 when the composite liquid hydrogen tanks catastrophically failed during testing¹⁹. This failure practically proved that the structural mass fractions required for SSTO flight were thermodynamically unachievable given the materials science of the era¹⁹.

Simultaneously, NASA developed the X-38 Crew Return Vehicle (CRV), intended to serve as a dedicated emergency lifeboat for the International Space Station. The X-38 returned to proven aerodynamics, modeling its shape directly on the Martin Marietta X-24A lifting body from the 1970s¹⁰. Unlike previous spacecraft, the X-38 was designed to be fully autonomous; it possessed no pilot control stick and relied entirely on satellite navigation to guide itself through atmospheric reentry²⁵. Because a lifting body can steer left or right of its ballistic path, it afforded a 1,300 km cross-range capability, allowing it to aim for safe terrestrial landing zones regardless of the ISS's orbital position¹⁰.

The X-38's terminal landing sequence was entirely novel. Rather than landing on a runway like the Space Shuttle, the X-38 deployed a massive 7,500-square-foot steerable parafoil at 20,000 feet¹⁰. The flight computers manipulated the parafoil's control lines, gliding the 25,000-pound craft to a precise landing on skid-plates at speeds below 40 mph¹⁰. Despite achieving stunning technical success in B-52 drop tests, and with the orbital V-201 prototype 90% complete, the \$500 million project was abruptly canceled in April 2002¹⁹. NASA Administrator Sean O'Keefe terminated the program as a cost-cutting measure amidst severe ISS budget overruns, forcing the United States to rely solely on Russian Soyuz capsules for emergency orbital return²⁶.

Part III: The Boeing X-37B Orbital Test Vehicle: Design and Engineering

Out of the ashes of these canceled megaprojects, the X-37 survived. Conceived by NASA in 1999 as part of the "Future-X" program to test reusable launch technologies, the project was contracted to Boeing's California-based Phantom Works division²⁸. Initially, NASA envisioned a vehicle that could be carried into orbit inside the payload bay of the Space Shuttle²⁹. However, following the 2003 Space Shuttle *Columbia* disaster, it was determined that risking a crewed Shuttle flight for an uncrewed payload was unacceptable, and the X-37 was reconfigured to launch vertically atop a Delta or Atlas rocket²⁹.

In September 2004, recognizing the vehicle's immense military utility and the need for a classified testbed, NASA transferred the program to the Defense Advanced Research Projects Agency (DARPA)²¹. Following a series of atmospheric drop tests from the Scaled Composites *White Knight* carrier aircraft in 2006, the USAF Rapid Capabilities Office assumed control of the orbital program, designating the operational vehicle the X-37B Orbital Test Vehicle (OTV)²¹. Since 2019, operational jurisdiction has resided with the United States Space Force (USSF), specifically managed by the 3rd Space Experimentation Squadron under Space Delta 9 at Schriever Space Force Base, Colorado²⁸.

Airframe and Hypergolic Propulsion

The X-37B is an approximately 120-percent-scale derivative of the earlier Boeing X-40A atmospheric test vehicle²⁹. It stands as the smallest and lightest orbital spaceplane ever flown²⁹. Measuring 29 feet 3 inches (8.8 meters) in length, with a wingspan of 14 feet 11 inches (4.5 meters) and a height of 9 feet 6 inches (2.9 meters), the vehicle is roughly one-quarter the size of the Space Shuttle³¹. It features a pickup-truck-sized payload bay measuring 7 by 4 feet, equipped with deployable gallium arsenide solar arrays and lithium-ion batteries to generate power for long-duration missions²¹. At launch, it weighs approximately 11,000 pounds (4,990 kg)³¹.

While conventional satellites rely on minor electrical thruster adjustments for station keeping, the X-37B is engineered for dynamic and extensive orbital maneuvering. Originally, NASA's design called for an Aerojet AR2-3 engine burning a mixture of hydrogen peroxide and JP-8 aviation fuel³⁷. However, under military stewardship, the X-37B was equipped with an Aerojet R-42 primary engine that utilizes a highly toxic hypergolic bipropellant mixture of nitrogen tetroxide and hydrazine³⁴.

This hypergolic system is incredibly reliable in the vacuum of space, as the chemicals spontaneously ignite upon contact without the need for complex, failure-prone ignition hardware³⁴. The propulsion system provides roughly 700 Newtons of thrust and an estimated total delta-v (change in velocity) of 3,100 meters per second³⁴. This affords the X-37B unprecedented maneuverability to drastically alter its orbit, change inclination planes, and evade adversary tracking networks³⁴. The trade-off for this reliability is the extreme toxicity of the hydrazine fuel. Upon returning to Earth, ground recovery crews at the Kennedy Space Center Orbiter Processing Facility (OPF) must wear fully encapsulated SCAPE (Self-Contained Atmospheric Protective Ensemble) hazmat suits when approaching the vehicle to safe the fuel lines and prevent fatal exposure to off-gassing fumes⁷.

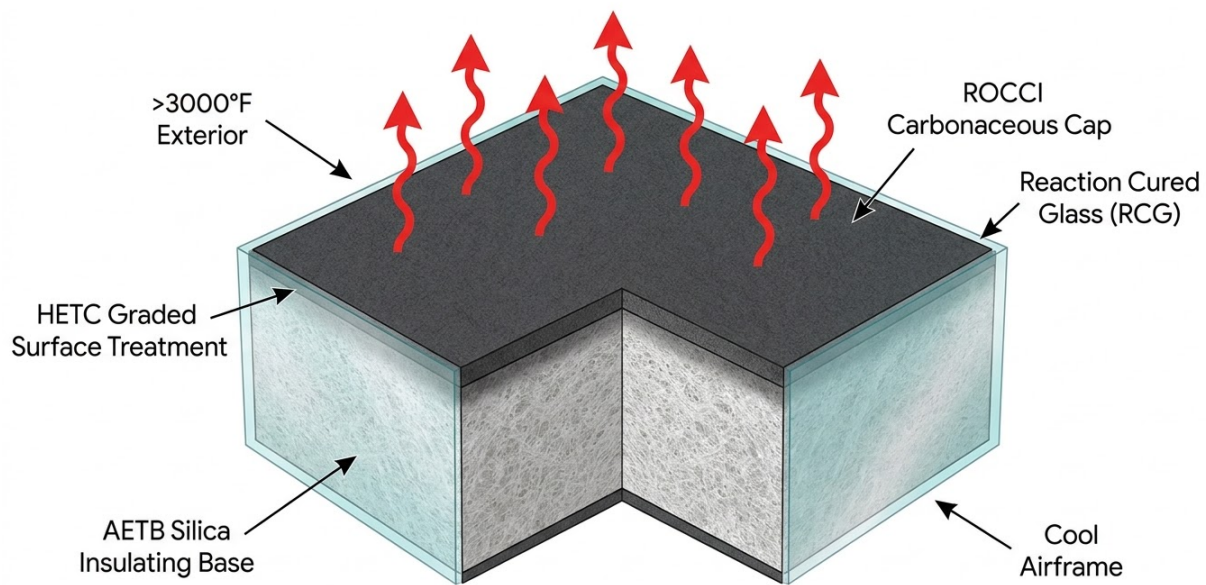
Advanced Thermal Protection System (TPS)

One of the most critical breakthroughs enabling the X-37B's rapid reusability is its thermal protection system. The Space Shuttle relied heavily on Reinforced Carbon-Carbon (RCC) for its highest-temperature leading edges and nose cone. While RCC was capable of withstanding 70 W/cm² of heat flux, it was exceptionally heavy (1.6 g/cm³), brittle, required 12-month manufacturing lead times, and cost upward of \$5 million per component⁴¹. Furthermore, its brittleness made it highly susceptible to catastrophic impact damage from orbital debris or foam shedding, as witnessed during the *Columbia* disaster.

To rectify these flaws, NASA Ames Research Center developed TUFROC (Toughened Uni-piece Fibrous Reinforced Oxidation-resistant Composite) specifically for the X-37B³⁴. TUFROC is a patented, highly engineered dual-layer composite system. The outer layer consists of a high-temperature carbonaceous cap (ROCCI) embedded with carbon fibers, designed to actively re-radiate extreme heat exceeding 3,000°F (1,650°C) back into the atmosphere⁴¹. This cap is adhesively and mechanically bonded via a graded surface treatment—which

accommodates differing rates of thermal expansion—to a highly porous, low-thermal-conductivity silica base known as AETB (Alumina-Enhanced Thermal Barrier)⁴¹. The entire assembly is then coated with Reaction Cured Glass (RCG) to prevent oxidation during the fiery descent⁴¹.

TUFROC provides equivalent or superior thermal protection to RCC but at a fraction of the mass (0.4 g/cm³) and cost (roughly \$50,000 per component), while offering significantly enhanced impact resistance⁴¹. The acreage of the vehicle's belly is protected by Toughened Uni-piece Fibrous Insulation (TUF) tiles, an evolution of Shuttle tile technology. Unlike the fragile glass coatings of the Shuttle era, TUF treatments penetrate deep into the silica substrate, creating a highly durable, water-resistant surface that mitigates the risk of moisture absorption prior to launch³⁴.



Part IV: Operational Flight History and Envelope Expansion

The US Space Force currently operates a fleet of two known X-37B vehicles, which have collectively flown eight classified missions²⁸. Originally designed with a baseline mission duration of 270 days, the program has systematically shattered its own endurance records, demonstrating escalating ambitions in autonomous operation, orbital manipulation, and payload complexity²⁹.

Mission Designation	Launch Date	Landing Date	Duration (Days)	Launch Vehicle	Notable Payloads & Milestones

OTV-1	Apr 22, 2010	Dec 3, 2010	224	Atlas V 501	First autonomous orbital runway landing by the US. Landed at Vandenberg AFB.
OTV-2	Mar 5, 2011	Jun 16, 2012	468	Atlas V 501	Maiden flight of Vehicle 2. Significantly exceeded 270-day design baseline.
OTV-3	Dec 11, 2012	Oct 17, 2014	674	Atlas V 501	Second flight of Vehicle 1. Confirmed vehicle reusability.
OTV-4	May 20, 2015	May 7, 2017	718	Atlas V 501	Tested Aerojet Rocketdyne XR-5A Hall-effect thruster. First landing at KSC.
OTV-5	Sep 7, 2017	Oct 27, 2019	780	Falcon 9	First launch on SpaceX Falcon 9. Higher inclination orbit.

					ASETS-II heat pipe test.
OTV-6	May 17, 2020	Nov 12, 2022	908	Atlas V 501	First use of aft service module. Deployed FalconSat-8 ; NRL solar power beaming test.
OTV-7	Dec 28, 2023	Mar 7, 2025	434	Falcon Heavy	First launch to Highly Elliptical Orbit. First successful aerobraking maneuver.
OTV-8	Aug 21, 2025	<i>In Orbit</i>	<i>Ongoing</i>	Falcon 9	Testing laser communications (via Starlink) and quantum inertial navigation sensors.

Early Endurance and Propulsion Testing (OTV-1 to OTV-5)

The inaugural flight, OTV-1, launched on an Atlas V 501 rocket in April 2010. It remained in Low Earth Orbit (LEO) for 224 days, successfully executing the first autonomous orbital runway landing by the United States at Vandenberg Air Force Base, sustaining only a minor tire blowout upon touchdown²¹. Subsequent missions methodically pushed the duration limits. OTV-2 lasted 468 days, and OTV-3 pushed to 674 days, irrevocably proving the long-term reliability of the platform's solar arrays, thermal control systems, and avionics²⁸.

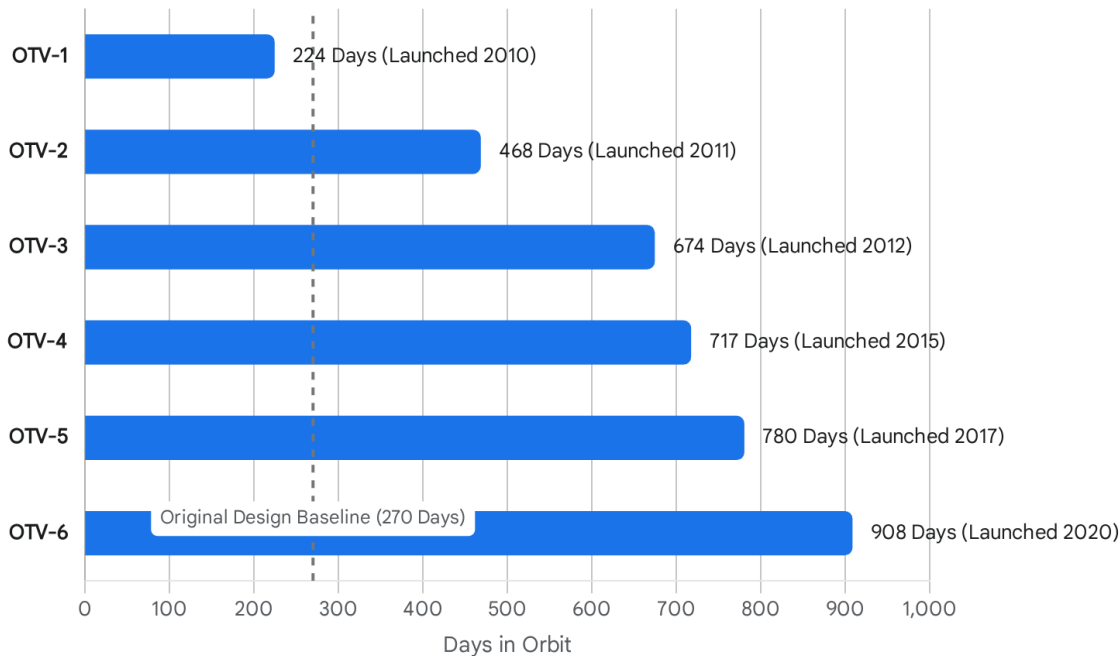
OTV-4 (718 days) significantly expanded the program's scope by serving as a functional testbed for external hardware. It tested an Aerojet Rocketdyne XR-5A Hall-effect electric propulsion thruster for the Advanced Extremely High Frequency (AEHF) military communications satellite program, alongside NASA experiments exposing materials and seeds to the space environment²⁹. OTV-5 marked a paradigm shift as it was the first mission to launch atop a SpaceX Falcon 9 rocket. During its 780 days in space, the vehicle utilized its hypergolic propulsion to actively alter its orbital inclination, heavily expanding its operational maneuvering envelope²⁸.

Modularity, Aerobraking, and Quantum Navigation (OTV-6 to OTV-8)

Launching in May 2020, OTV-6 introduced a bolted-on service module ring attached to the aft of the vehicle, vastly expanding payload capacity²⁹. During its record-breaking 908-day mission, it deployed the FalconSat-8 small satellite (developed by USAF Academy cadets) and hosted a Naval Research Laboratory (NRL) experiment converting solar energy into radio frequency microwaves—a critical precursor to space-based solar power beaming²⁹.

X-37B Orbital Endurance by Mission

■ Days in Orbit (U.S. Space Force / USAF)



Since its inaugural flight in 2010, the X-37B program has systematically expanded its operational endurance. Originally designed for a 270-day baseline, the spaceplane has proven capable of sustaining autonomous operations for over two and a half years.

Data sources: [EBSCO](#), [Wikipedia](#), [19FortyFive](#)

The two most recent missions represent a profound evolution from mere endurance testing to aggressive operational agility. OTV-7 (USSF-52) launched in December 2023 atop a massive SpaceX Falcon Heavy rocket, which injected the spaceplane into a highly elliptical, high-Earth orbit (HEO)—a first for the program²⁸. In late 2024, the Space Force announced that OTV-7 would execute a novel "aerobraking" maneuver. Aerobraking utilizes the aerodynamic drag of the Earth's upper atmosphere to decelerate the spacecraft during the perigee (closest approach) of an elliptical orbit⁴⁶. Over multiple passes, this friction bleeds off velocity, circularizing and lowering the orbit while conserving vast amounts of hypergolic propellant⁴⁷. This maneuver was executed to safely dispose of the service module in compliance with space debris mitigation standards before OTV-7 successfully landed at Vandenberg in March 2025⁴⁶. Breaking the trend of extended endurance, the Space Force executed an extraordinarily rapid turnaround. Less than six months later, on August 21, 2025, OTV-8 launched on a Falcon 9⁴⁵. For the first time, Space Force officials were highly transparent about the payload. OTV-8 is testing a high-bandwidth inter-satellite laser communications system designed to link directly

with proliferated commercial constellations in LEO (such as SpaceX's Starlink)⁵². Laser communications offer immense data throughput and are highly resistant to enemy jamming or interception⁵².

Furthermore, OTV-8 carries a state-of-the-art quantum inertial sensor. By measuring the interference patterns of rotated and accelerated atoms, the sensor allows the spacecraft to calculate its exact position and orientation without relying on external GPS signals⁵². This ensures robust maneuverability in GPS-denied environments during a potential conflict, directly supporting the mandates of the Fifth Space Operations Squadron⁵².

Part V: Dissecting the Shadows: Doctrine and the "Orbital Bomber" Fallacy

The intense classification of the X-37B's payloads, combined with its ability to change orbits and disappear from civilian tracking networks, has catalyzed rampant global speculation. Russian and Chinese officials, as well as fringe theorists, have repeatedly accused the United States of fielding a covert orbital bomber, a space-based kinetic weapon, or an anti-satellite (ASAT) interceptor²⁸.

The Kinetic Physics Limitations

From a perspective of pure physics and orbital mechanics, the X-37B is an exceptionally poor weapons platform. As detailed in the Secure World Foundation's (SWF) *Global Counterspace Capabilities Report*, the vehicle's 7-by-4-foot payload bay is far too small to carry significant munitions or nuclear payloads³⁵.

Furthermore, a spaceplane acting as an orbital bomber faces severe kinetic disadvantages. Dropping a munition from orbit is not akin to dropping a bomb from an airplane; the munition shares the vehicle's horizontal velocity of approximately 17,500 mph²⁴. It requires a massive retrograde burn to shed that velocity and re-enter the atmosphere. By the time a weapon decelerates and navigates reentry, its trajectory is highly predictable, and the time-to-target is far slower than a terrestrial Intercontinental Ballistic Missile (ICBM)⁵⁵. Additionally, the X-37B relies on fixed terrestrial runways for landing and requires a lengthy pre-launch integration period atop a heavy-lift rocket, making the system highly vulnerable to preemptive terrestrial strikes³¹. Civilian and adversary Space Situational Awareness (SSA) networks can easily track its position; an unpowered glider re-entering the atmosphere cannot "sneak up" on a terrestrial target³¹.

The Doctrine of Dynamic Space Operations (DSO)

The true strategic value of the X-37B lies not in kinetic strikes, but in information dominance, technological prototyping, and operational unpredictability. Traditional satellites operate in predictable Keplerian orbits. Once an adversary tracks a satellite, they can accurately predict its future location, making it highly vulnerable to ASAT missiles, dazzling lasers, or grappling spacecraft³³.

The X-37B enables a revolutionary USSF doctrine articulated by Chief of Space Operations Gen. B. Chance Saltzman known as "Dynamic Space Operations" (DSO)⁴⁹. DSO envisions a space architecture characterized by continuous orbital maneuvering. By utilizing its hypergolic engines and demonstrating fuel-saving techniques like aerobraking, the X-37B proves that a spacecraft can constantly alter its orbital parameters, continuously changing its inclination and perigee to frustrate adversary targeting³⁵. An adversary calculating an intercept solution based on yesterday's orbital data will find empty space today.

Moreover, the X-37B serves as an unassailable engineering testbed. The Space Force can deploy prototype quantum sensors, optical reconnaissance payloads, or signals intelligence (SIGINT) equipment, expose them to the harsh radiation and thermal extremes of the space environment, and then return them to Earth for meticulous forensic engineering analysis. This iterative process drastically accelerates the deployment of next-generation defense systems, saving billions in potential in-orbit failures²¹.

Part VI: Global Proliferation and Contemporary Spaceplanes

The strategic advantages demonstrated by the X-37B have not gone unnoticed, sparking a global pursuit of reusable orbital and suborbital technologies across both adversarial militaries and commercial enterprises.

China's Shenlong and Rendezvous Operations

The most direct strategic competitor to the X-37B is China's highly secretive experimental spaceplane, informally known as *Shenlong* (Divine Dragon). Developed by the state-owned China Academy of Launch Vehicle Technology (CALT), Shenlong is launched vertically atop a crew-rated Long March 2F rocket⁶³.

Shenlong has executed multiple orbital missions since 2020, demonstrating an alarming proficiency in Rendezvous and Proximity Operations (RPO). During its missions, commercial SSA providers like LeoLabs, alongside independent trackers such as Marco Langbroek and Scott Tilley, have observed the spacecraft releasing multiple unidentified objects into orbit⁶³. These objects, sometimes referred to as "wingmen," have been detected emitting specific radio signals near tracking stations, suggesting they act as data relays, inspector satellites, or subsatellites designed to practice capturing or interacting with other orbital assets⁶³.

Victoria Samson of the Secure World Foundation notes that such RPO capabilities inherently carry deep dual-use applications; a satellite capable of inspecting or refueling a friendly asset possesses the exact kinetic maneuverability necessary to grapple, dazzle, or physically disable an enemy asset⁶⁴.

Commercial and Civilian Architectures

Beyond military applications, the commercial sector is rapidly advancing spaceplane technology to capitalize on the burgeoning Low Earth Orbit economy, seeking cheaper and more rapid access to space.

- **Dream Chaser:** Developed by Sierra Space, the Dream Chaser is the direct commercial beneficiary of the BOR-4 and HL-20 aerodynamic lineage². Designed primarily for cargo resupply to the ISS under NASA commercial contracts, its lifting-body design allows for a gentle, 1.5-G reentry². This low-G profile is crucial for returning sensitive biological experiments or fragile manufactured goods from orbit, and its cross-range maneuverability enables it to land at conventional commercial airports².
- **Space Rider:** The European Space Agency (ESA) is developing the Space Rider, an uncrewed robotic laboratory designed to launch on the Vega-C rocket. Capable of carrying 600 kg of cargo for two-month missions, it aims to provide Europe with independent, reusable microgravity research capabilities, utilizing a parafoil landing system reminiscent of the canceled X-38⁶⁹.
- **Dawn Aerospace Mk-II Aurora:** Operating in the suborbital regime, New Zealand-based Dawn Aerospace is developing the Mk-II Aurora, a rapidly reusable spaceplane designed for high-frequency suborbital flights, taking off and landing conventionally from standard airports to conduct atmospheric research and microgravity testing⁷³.
- **Radian One:** Radian Aerospace is aggressively attempting to resurrect the Single-Stage-To-Orbit (SSTO) dream with the *Radian One* spaceplane. To overcome the tyrannical constraints of the rocket equation that doomed the X-33, Radian employs a unique "stage zero" rocket sled assist²³. This ground-based sled accelerates the spaceplane along a rail to supersonic speeds before the vehicle fires its internal methalox engines, preserving vast amounts of onboard fuel for the orbital climb²⁴. As of late 2024, Radian successfully conducted low-speed taxi tests (reaching 50 knots) of its PFV01 subscale prototype in Abu Dhabi. The company aims for the vehicle to carry 2 to 5 crew members with an unprecedented 48-hour turnaround time between flights²³.

Part VII: The Legal and Regulatory Vanguard

The rapid deployment of maneuvering spaceplanes, the unannounced release of "wingmen" subsatellites, and the proliferation of RPO technologies operate in the deeply ambiguous margins of international space law, threatening to severely destabilize the orbital environment.

The Outer Space Treaty and the Ambiguity of Weaponization

The foundation of modern space governance is the 1967 Outer Space Treaty (OST). Article IV of the OST explicitly prohibits the placement of nuclear weapons or other weapons of mass destruction (WMDs) in Earth orbit, or the installation of such weapons on celestial bodies⁶¹. However, the OST is fundamentally not a comprehensive demilitarization treaty. It contains no prohibition against the placement of conventional weapons in Earth orbit, nor does it forbid military reconnaissance, secure communications networks, or technology testbeds like the X-37B⁵⁹.

This legal vacuum has allowed the militarization of space to proceed unchecked. As nations develop "bodyguard satellites" designed to protect larger assets and spaceplanes capable of sophisticated RPO, the distinction between a peaceful satellite servicer and a conventional

space weapon becomes entirely a matter of intent, which is legally unprovable until a kinetic attack occurs⁵⁹. Furthermore, the OST lacks any binding verification mechanism; there are no treaty-mandated orbital inspections. Compliance verification relies entirely on National Technical Means (NTM) of surveillance, meaning that ascertaining whether a classified payload bay holds a scientific experiment or a conventional weapon is nearly impossible from the ground⁵⁸.

The Registration Convention and Space Traffic Management

To establish liability and transparency, the 1975 Registration Convention (expanding upon Article XI of the OST) requires launching states to record the basic orbital parameters, general function, and launch date of space objects in a United Nations registry maintained by the Office for Outer Space Affairs (UNOOSA)⁸¹.

In practice, compliance with the Registration Convention is woefully inadequate. States frequently delay registration for months or even years, and the functional descriptions provided are often deliberately vague, utilizing boilerplate terms like "technology demonstration" or "scientific research"⁸⁴. The operations of vehicles like the X-37B and Shenlong heavily exacerbate these transparency failures. When Shenlong ejects multiple subsatellites into orbit, China rarely registers them, leaving civilian operators and rival militaries guessing as to their orbital trajectories, mass, and intent⁶³.

As the orbital domain becomes increasingly congested with commercial mega-constellations like Starlink (comprising thousands of active satellites), the lack of strict Space Traffic Management (STM) rules, combined with highly maneuverable, unannounced military spacecraft, dramatically increases the risk of accidental collisions or miscalculated geopolitical escalations⁸⁶. Without a federated governance model that mandates standardized collision avoidance thresholds and transparent maneuver notifications, the orbital environment remains highly precarious⁸⁸.

Conclusion

The Boeing X-37B is not merely a technological marvel; it is the physical manifestation of a profound strategic pivot in aerospace doctrine. By proving that a spacecraft can launch, endure the extreme thermal and radioactive realities of the space vacuum for over two and a half years, maneuver dynamically through the atmosphere, and return to Earth autonomously to be flown again, the X-37B has shattered the paradigm of expendable space architecture.

It stands firmly on the shoulders of the Cold War engineers who dreamt of the Dyna-Soar and meticulously flew the lifting bodies of the 1960s and 1970s. Today, equipped with quantum inertial sensors that negate the need for GPS, laser communications that bypass traditional radio frequency vulnerabilities, and the aerodynamic capability to execute fuel-saving aerobraking maneuvers, it provides the United States with unparalleled operational resilience and rapid technological prototyping capabilities.

However, its success has undeniably catalyzed a global response. As China's Shenlong expands its operational envelope with unannounced subsatellite deployments, and commercial entities race to field their own fleets of reusable lifting bodies and SSTO platforms, the orbital domain is rapidly transitioning into a highly contested and dangerously congested environment. Moving forward, the true challenge facing the international community will not be mastering the

thermodynamics of hypersonic reentry or the physics of quantum navigation, but rather forging the robust legal frameworks and space traffic management protocols necessary to ensure that the era of the spaceplane does not inadvertently trigger an era of catastrophic orbital conflict.

Works cited

1. Design Evolution and AHP-based Historiography of Lifting Reentry, https://ideas.esa.int/core/apps/IMT/UploadedFiles/00/f_16b71f59009dbdb1670b31499300a9df/Design_Evolution_and_AHP-based_Historiography_of_Lifting_Ree.pdf?v=1788653130
2. The Other Guys - Smithsonian Magazine, <https://www.smithsonianmag.com/air-space-magazine/the-other-guys-11799113/>
3. Heavenly Bodies – Part I - aircscape Magazine, <https://airscapemag.com/2017/10/14/heavenly-bodies-part-i/>
4. X-20 Dyna-Soar Spaceplane Was Decades Ahead of Its Time, <https://www.defensemecanetwork.com/stories/what-might-have-been-x-20-dyna-soar/>
5. Preliminary Structural Evaluation and Design of the Dream Chaser, https://www.academia.edu/37124414/Preliminary_Structural_Evaluation_and_Design_of_the_Dream_Chaser_Space_System_lifting_body_concept
6. Spaceplane - Wikipedia, <https://en.wikipedia.org/wiki/Spaceplane>
7. A History of the US Air Force Space Launch Enterprise, 1947–2020, <https://www.govinfo.gov/content/pkg/GOVPUB-D301-PURL-gpo183897/pdf/GOVPUB-D301-PURL-gpo183897.pdf>
8. Wingless Flight - The Lifting Body Story - NASA, <https://www.nasa.gov/wp-content/uploads/2023/03/sp-4220.pdf?emrc=ef9cda>
9. Northrop HL-10 - Military Wiki | Fandom, https://military-history.fandom.com/wiki/Northrop_HL-10
10. The X-38: NASA's Unfinished Space Station Lifeboat, <https://newspaceconomy.ca/2025/11/04/the-x-38-nasas-unfinished-space-station-lifeboat/>
11. hl-20 lifting body: Topics by Science.gov, <https://www.science.gov/topicpages/h/hl-20+lifting+body>
12. Mikoyan-Gurevich MiG-105 - Wikipedia, https://en.wikipedia.org/wiki/Mikoyan-Gurevich_MiG-105
13. Dream Chaser, The Vehicle Always Just About Ready To Fly, <https://talkoftitusville.com/2026/07/12/dream-chaser-the-vehicle-always-just-about-ready-to-fly/>
14. On the Feasibility of Landing the Dream Chaser Space Vehicle in, https://www.researchgate.net/profile/Luke-Colvin/publication/335977940_On_the_Feasibility_of_Landing_the_Dream_Chaser_Space_Vehicle_in_South_Africa/links/5d889980458515cbd1b50edb/On-the-Feasibility-of-Landing-the-Dream-Chaser-Space-Vehicle-in-South-Africa.pdf
15. The HL-20: NASA's Unbuilt Lifting Body Spaceplane, <https://newspaceconomy.ca/2025/11/04/the-hl-20-nasas-unbuilt-lifting-body-spac>

- [eplane/](#)
16. NASA's New Cargo Vehicle For The International Space Station, <https://www.robertnovell.com/nasas-new-cargo-vehicle-for-the-international-space-station-was-first-known-as-the-mig-105-january-29-2016-2-2/>
 17. Russian Spaceplanes, <https://weebau.com/rockwing/rusplanes.htm>
 18. Spaceplanes | Chutes.nl, http://chutes.nl/missions/earth_missions/spaceplanes.html
 19. Unflown NASA Concepts, <https://www.nasaemployees.com/unflown-nasa-concepts>
 20. Paper Session III-C - A New "Old Way" to Space - Scholarly Commons, <https://commons.erau.edu/cgi/viewcontent.cgi?article=1131&context=space-congress-proceedings>
 21. The Pentagon's Mystery Spaceplane Takes Off - KeepTrack.space, <https://keeptrack.space/today-in-space-history/today-in-space-history-2026-04-22>
 22. r/spacex on Reddit: "Steps toward building the first orbital passenger", https://www.reddit.com/r/spacex/comments/4gaocs/steps_toward_building_the_first_orbital_passenger/
 23. Radian One, A New Spaceplane Is Taking Shape - Space Voyaging, <https://www.spacevoyaging.com/news/2024/04/07/radian-one-a-new-spaceplane-is-taking-shape/>
 24. Ground Tests Are Go: What To Know About Radian's Prototype, <https://simpleflying.com/raidan-prototype-space-plane/>
 25. NASA X-38 - Photos & Videos - NET-MAQUETTES, <https://www.net-maquettes.com/pictures/nasa-x-38/>
 26. NASA X-38 - Wikipedia, https://en.wikipedia.org/wiki/NASA_X-38
 27. X-38, <http://www.astronautix.com/x/x-38.html>
 28. Boeing X-37 | Engineering | Research Starters - EBSCO, <https://www.ebsco.com/research-starters/engineering/boeing-x-37/>
 29. Boeing X-37 - Wikipedia, https://en.wikipedia.org/wiki/Boeing_X-37
 30. X-37B: America's Spaceplane - Grey Dynamics, <https://greydynamics.com/x-37b-americas-spaceplane/>
 31. X-37B: The Air Force's Mysterious Space Plane, <https://www.space.com/25275-x37b-space-plane.html>
 32. L'X-37 è uno spaziplano sperimentale non pilotato sviluppato per, <https://svppbellum.blogspot.com/2020/09/lx-37-e-uno-spaziplano-sperimentale.html>
 33. World: Space | Air & Space Forces Magazine, <https://www.airandspaceforces.com/article/world-space-2/>
 34. X-37B: America's Secret Space plane - A Continual Learner, <https://acontinuallearner.medium.com/x-37b-americas-secret-space-plane-fd61b331d998>
 35. X-37B: The Space Force's Mystery Spaceplane - MiGFlug, <https://migflug.com/afterburner/x-37b-space-force-mystery-spaceplane/>
 36. X-37B Orbital Test Vehicle | Air & Space Forces Magazine, <https://www.airandspaceforces.com/weapons/x-37b/>

37. BOEING X-37B (OTV-1) by RADMRockstone on DeviantArt,
<https://www.deviantart.com/radmrockstone/art/BOEING-X-37B-OTV-1-938669840>
38. Why is the X-37B's large nozzle offset to right from the spacecraft's,
<https://space.stackexchange.com/questions/21423/why-is-the-x-37bs-large-nozzle-offset-to-right-from-the-spacecrafts-mid-plane>
39. SpaceX launched X-37B space plane landed today after over 2,
https://www.reddit.com/r/spacex/comments/dny511/spacex_launched_x37b_space_plane_landed_today/
40. Chronology of KSC and KSC Related Events for 2012,
<https://ntrs.nasa.gov/api/citations/20130011640/downloads/20130011640.pdf?attachment=true>
41. Advanced Lightweight TUFROC Thermal Protection System for,
<https://ntrs.nasa.gov/api/citations/20240002574/downloads/2574-TM-TUFROC-2024.pdf?attachment=true>
42. Star power! How glass and ceramics push us deeper into space,
<https://bulletin.ceramics.org/article/star-power-how-glass-and-ceramics-push-us-deeper-into-space/>
43. Reusable Thermal Protection Systems | PDF - Scribd,
<https://www.scribd.com/document/916957034/20160003291-Presentation-NASA>
44. (PDF) Durability Assessment of Tile-Type Reusable Thermal,
https://www.researchgate.net/publication/399700999_Durability_Assessment_of_Tile-Type_Reusable_Thermal_Protection_Materials
45. The X-37B Just Passed a Year in Orbit on Its Eighth Secret Mission,
<https://www.19fortyfive.com/2026/08/the-x-37b-just-passed-a-year-in-orbit-on-its-eighth-secret-mission-russia-calls-it-a-nuclear-bomber/>
46. X-37B Orbital Test Vehicle concludes seventh successful mission,
<https://www.spaceforce.com/news-events/article/x-37b-orbital-test-vehicle-concludes-seventh-successful-mission>
47. U.S. Space Force Will Use Aerobraking to Change X-37B Orbit,
<https://theaviationist.com/2024/10/10/aerobraking-to-change-x-37b-orbit/>
48. X-37B begins novel space maneuver,
<https://www.spaceforce.com/news-events/x-37b-begins-novel-space-maneuver>
49. Space Force X-37 Aerobraking Might Inform Future Satellite Design,
<https://www.airandspaceforces.com/space-force-x-37-aerobraking/>
50. The Secretive Spaceplane of the U.S. Space Force Conducts First-of,
<https://www.smithsonianmag.com/smart-news/the-secretive-spaceplane-of-the-us-space-force-conducts-first-of-its-kind-maneuvers-180985425/>
51. Secretive US Space Force X-37B space plane 'breaks new ground',
<https://www.space.com/space-exploration/launches-spacecraft/secretive-us-space-force-x-37b-space-plane-breaks-new-ground-as-it-returns-to-earth-after-434-days-in-orbit-photos>
52. The X-37B's Next Leap: How Quantum Sensors and Laser Links,
<https://modernengineeringmarvels.com/2025/07/31/the-x-37bs-next-leap-how-quantum-sensors-and-laser-links-could-reshape-space-operations/>
53. US Military's Robotic X-37B Spacecraft on Secret Space Mission,

- <https://www.ir-ia.com/US-Robotic-X-37B-Spacecraft.html>
54. Boeing-Built X-37B Spaceplane Set for Eighth Mission - Stock Titan,
<https://www.stocktitan.net/news/BA/boeing-built-x-37b-spaceplane-set-for-eighth-t-wckxqtfu49r.html>
 55. X-37B Orbital Test Vehicle Fact Sheet - Secure World Foundation,
<https://www.swfound.org/publications-and-reports/x-37b-orbital-test-vehicle-fact-sheet>
 56. Reports - Secure World Foundation,
https://www.swfound.org/reports?2c0f6f25_page=4
 57. Spaceplanes: why we need them, why they have failed, and how,
<https://www.thespacereview.com/article/4791/1>
 58. Verifying the Prohibition on the Placement of Nuclear Weapons in,
<https://www.gcsp.ch/news/verifying-prohibition-placement-nuclear-weapons-outer-space>
 59. US confirms weapons deployed in orbit: Is new global space war about to begin?,
<https://zeenews.india.com/world/us-confirms-weapons-deployed-in-orbit-is-new-global-space-war-about-to-begin-3071767.html>
 60. Dynamic Orbital Manoeuvre Revolution,
<https://www.orbitalwarfarehistory.com/dynamic-orbital-manoevre-revolution>
 61. U.S. confirms it has weapons in orbit | Astronomy.com,
<https://www.astronomy.com/space-exploration/u-s-confirms-it-has-weapons-in-orbit/>
 62. 2026 Global Counterspace Capabilities Report,
<https://www.swfound.org/publications-and-reports/2026-global-counterspace-capabilities-report>
 63. Secret Shenlong spaceplane launched a mysterious object into space,
<https://universemagazine.com/en/secret-shenlong-spaceplane-launched-a-mysterious-object-into-space/>
 64. Chinese Spacecraft Has Secret Military Use, Experts Say - Futurism,
<https://futurism.com/the-byte/chinese-spacecraft-secret-military>
 65. Mysterious China Space Plane: Could it Jam Santa Claus Trajectory?,
<https://www.leonarddavid.com/mysterious-china-space-plane-could-it-jam-santa-clause-trajectory/>
 66. 2023 December - Leonard David's INSIDE OUTER SPACE,
<https://www.leonarddavid.com/2023/12/>
 67. GLOBAL COUNTERSPACE CAPABILITIES,
https://cdn.prod.website-files.com/66dcc6872f6ed23bce1db235/69d5402700ec843d95073a1e_SWF_Global_Counterspace_Capabilities_2026.pdf
 68. SWF's 2023 fact sheets on anti-satellite testing, military ... - SatNews,
<https://satnews.com/2023/07/17/swfs-2023-fact-sheets-on-anti-satellite-testing-military-and-intelligence-rpos-and-the-x-37b/>
 69. Space Rider - Space Launch Schedule,
<https://www.spacelaunchschedule.com/spacecraft/space-rider/>
 70. Space Rider - Telespazio,
<https://www.telespazio.com/en/business/space-programmes/space-rider>

71. Vega (rocket) - Wikipedia, [https://en.wikipedia.org/wiki/Vega_\(rocket\)](https://en.wikipedia.org/wiki/Vega_(rocket))
72. Experimental Spaceplanes - Chutes.nl, http://chutes.nl/missions/testmissions/entry_test/experimental-spaceplanes.html
73. Meet Dream Chaser, The Next-Generation Space Plane - YouTube, <https://www.youtube.com/watch?v=rqdmпуwOw7g>
74. Dawn Aerospace - Wikipedia, https://en.wikipedia.org/wiki/Dawn_Aerospace
75. Radian Aerospace Deep Dive - Payload Space, <https://payloadspace.com/radian-aerospace-deep-dive/>
76. The Next-Generation of Single-Stage-to-Orbit Spaceplanes ... - Recall, <https://www.recall.it/summary/astronomy/the-next-generation-of-single-stage-to-orbit-spaceplanes-with-radians-richard-humphrey-or-e2022>
77. Radian Aerospace tests prototype for space plane - Cosmic Log, <https://cosmiclog.com/2024/09/25/radian-aerospace-tests-prototype-for-space-plane/>
78. Radian Aerospace Reveals First Prototype Flight Vehicle, <https://www.prnewswire.com/news-releases/radian-aerospace-reveals-first-prototype-flight-vehicle-completes-initial-round-of-taxi-tests-302257435.html>
79. International Legal Agreements Relevant to Space Weapons, <https://www.ucs.org/resources/legal-agreements-space-weapons>
80. Orbital Militarization and the Weaponization of Space: Legal and, <https://www.diplomacyandlaw.com/post/orbital-militarization-and-the-weaponization-of-space-legal-and-strategic-limits>
81. An Expert Legal Analysis of Article XI of the Outer Space Treaty, <https://kwasuspace.kwasu.edu.ng/bitstreams/27814e6b-bb35-4b52-aeb5-4665915cff41/download>
82. Minimum International Norms For Managing Space Traffic, Space, <https://scholar.smu.edu/cgi/viewcontent.cgi?article=4101&context=jalc>
83. International Law and the Regulation of Outer Space, <https://www.diplomacyandlaw.com/post/international-law-and-the-regulation-of-outer-space>
84. Critical registrations along the satellite lifecycle - ITU, <https://www.itu.int/hub/2025/10/critical-registrations-along-the-satellite-lifecycle/>
85. Measuring Compliance to the United Nations Registration Convention, https://www.researchgate.net/publication/404714524_Measuring_Compliance_to_the_United_Nations_Registration_Convention
86. Advancements in Space Law: Satellite Communications Industry, <https://www.cambridge.org/core/journals/international-journal-of-legal-information/article/advancements-in-space-law-satellite-communications-industry-regulations-and-obligations-for-orbital-debris-mitigation/AFDF40D53A7F1FF228BF1AA466E8A24D>
87. Governing Outer Space: A Conference of the Parties for the Outer, <https://www.belfercenter.org/research-analysis/space-cop-governance>
88. Responsible Rules in Orbit: Federated Governance for a Crowded, <https://blog.neuraspace.com/ceo-corner/responsible-rules-in-orbit-federated-governance-for-a-crowded-and-congested-space>